Reg No.:_______________ Name:__________________________

1100EET307122103

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: EET307

Course Name: SYNCHRONOUS AND INDUCTION MACHINES

(GRAPH SHEET NEEDED)

Max. Marks: 100 Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

			Marks
1		Discuss about the different types of windings in alternator	3
2		What are the causes of harmonics in the induced voltage of an alternator and how its effects are minimised?	3
3		Draw and explain the phasor diagram of a salient pole alternator supplying to a lagging power factor load	3
4		Enumerate the requirement of proper parallel operation of alternators	3
5		Discuss about any two starting methods of synchronous motors	3
6		Derive expressions for starting torque in three phase induction motor	3
7		Explain about working of star delta starter for three phase induction Motor	3
8		Explain about V/f speed control method in three phase induction motors	3
9		Compare the operation of induction Generator in grid connected mode and self excited mode	3
10		Explain about working of split phase induction motor	3

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

11	a)	List the advantages of using stationary armature in alternator	7
	b)	A 3 phase alternator,50 Hz,8 pole, has 180 conductors per phase ,flux/pole is .054wb,find the induced line emf (assume $k_p=k_d=1$)	7
12	a)	What is the significance of winding factor in the induced voltage of induction motor? Derive an expression for the same	7
	b)	Explain different types of alternators based on rotor construction	7

Module -2

13	a)	Find the full load regulation by MMF method of a 100 kVA , 2kV 3 phase 50 Hz star connected alternator at 0.8 power factor lag having the following test data, armature resistance= 0.2Ωper phase	14

If in A	10	20	25	30	40	50
Vt (line voltage)	800	1500	1760	2000	2350	2600
Isc in A	-	20	25	30	-	-

14	a)	With circuit diagram explain about synchronisation of alternator using bright lamp method	7
	b)	Explain slip test and calculation of Xd and Xq	7

Module -3

15	a)	A 75Kw,400V,4 pole ,3 phase star connected synchronous motor has (0.04+j0.4)/phase ,compute for full load 0.8pf lead, armature current and gross mechanical power developed assuming an efficiency of 92.5%	7
	b)	Draw & explain about the power- angle characteristics of synchronous motor	7
16	a)	Discuss power stage diagram of induction motor	4
	b)	The power input of a 500V,50Hz, 6 pole, 3 phase induction motor running at 975 rpm is 40KW,the stator losses is 1 KW and friction and windage losses are 2KW.find slip,rotor copper losses,BHP and efficiency	10

Module -4

17	a)	Draw the circle diagram of a 20 HP,400V,50Hz,3 phase star connected induction motor from the following test data (line values) No load test : 400V 9A pf 0.2 Blocked rotor test : 200V 50A pf 0.4 from circle diagram find line current and power factor at full load and maximum power output	14
18	a)	Explain about different braking methods of induction motor	7
	b)	Explain how high starting torque and good operating efficiency is achieved in double cage induction motor	7

Module -5

19	a)	Explain why single phase induction motor is not self starting	7
	b)	Draw and discuss about speed - torque characteristics of induction generator	7
20	a)	Explain about induction Generator	7
	b)	Explain capacitor start capacitor run single phase induction motors	7